

MY PCC – Mobile Application & Admin Ecosystem

Supporting Document for AntiGravity Submission

Academic Year: 2025-2026

Abstract

MY PCC is a full-stack digital transformation system designed to modernize church operations. It includes a mobile application and centralized admin dashboard for managing digital resources, subscriptions, and financial tracking.

1. Introduction

The system digitizes church diary, hymn book, devotional materials, and monthly publications. It provides secure global access and centralized administration.

2. Objectives

- Digitize church resources
- Provide secure mobile access
- Enable subscription-based revenue model
- Ensure financial transparency
- Reduce administrative workload
- Build scalable cloud-ready architecture

3. System Architecture

Architecture Flow: Mobile App → Admin Dashboard → REST API → Database → Cloud Infrastructure.

4. Technology Stack

- Backend: Spring Boot
- Web Dashboard: Angular
- Mobile Application: Flutter
- Database: MySQL
- Security: JWT + Spring Security
- Cloud Hosting: AWS

5. System Modules

- User Management Module (Registration, Login, Role-based Access)
- Digital Resource Module (Diary, Hymn Book, Devotionals)
- Subscription & Payment Module (Online Payments, Expiry Tracking)
- Admin Dashboard Module (Upload Content, Manage Users, Reports)

6. Database Design

- users
- subscriptions
- hymns
- diary_readings
- payments
- magazines

7. Security Implementation

The system uses JWT authentication, password hashing, role-based authorization, and secure REST endpoints over HTTPS.

8. Deployment Architecture

Backend hosted on AWS EC2, Database on AWS RDS, File storage on S3, Content delivery via CloudFront, and access managed using IAM.

9. Conclusion

MY PCC is a scalable, secure, and cloud-ready digital transformation solution designed to enhance church operations and global member engagement.